

Five Layers of Obstruction: A Successor Architecture for Viability under Incomplete Observation

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Abstract

The obstruction programme has produced a calculus of certificates for viability under incomplete observation, a continuous-to-finite bridge, a selector unification, and an exact stochastic theory. This paper assembles them into the architecture their successor requires: five layers of obstruction, ordered by the resource whose absence each detects — dynamics (full-information exit), instantaneous belief-action compatibility, timing, certification, and policy class — each layer carrying the certificate families of the companion papers as its witnesses. Two generalizations are absorbed and carried to exact instances. General information structures: the selector framework is formulated on arbitrary observation partitions with read-then-act semantics, viability is monotone under refinement, and a two-patch instance shows the lattice matters — two incomparable structures receive opposite verdicts on the same belief, and refinement strictly erodes the certificates (twenty-eight versus twenty-six viable pairs) precisely as the bridge’s erosion identity predicts. Exit-time value functions: the guaranteed survival time is the quantity the first and third layers measure, its level sets are exactly the viability kernels, the timing certificate is the comparison of that value against the observation deadline, and on the audited classes the level-set identities and the bridge’s linear-programming thresholds coincide cell by cell. Every structural claim is verified in exact rational arithmetic on the companions’ worked systems, with one instance at each layer boundary in which the layer’s obstruction fires while every coarser layer is silent. The architecture is a presentation discipline, not a new mechanism; the strongest claims remain the companions’, each at the precision its development supports.

1. Introduction

Four papers form the obstruction programme: the calculus of obstruction certificates (Abaee, 2026a), the continuous-to-finite bridge for recourse-aware nonviability (Abaee, 2026b), the viable-selector unification (Abaee, 2026c), and the stochastic selector (Abaee, 2026d). This paper is the successor architecture they were building toward: the five-layer organization of obstruction by the resource whose absence each layer detects, with the two deferred generalizations — general information structures and exit-time value functions — absorbed and carried to exact, verified

instances on the programme’s worked systems.

The layers, top to bottom:

- **Layer 1 (dynamics).** Full-information exit: a drift or comparison condition under which the disturbance forces exit in bounded time against every control. No information-theoretic content; the base obstruction of Abaee 2026a.
- **Layer 2 (belief action).** One-step incompatibility across an information state: empty common admissible, safe, or tube-safe response sets; the epistemic-emptiness construction; the common-action certificate and its Farkas and Helly witnesses.
- **Layer 3 (timing).** The blind window: information that arrives after the guaranteed survival time has expired; the delayed-information bound, the guaranteed survival time σ^* , and the bridge’s finite linear programs with Farkas duals.
- **Layer 4 (certification).** The fibre criterion: no exact observation-only certifier when a fibre crosses the safe-set boundary; the certainly-safe relaxation as the maximal sound labelling.
- **Layer 5 (policy class).** Obstructions created by restricting the controller class — the certainty-equivalence trap — which can operate under injective observation with every coarser layer silent.

The architecture earns its keep at the boundaries, and Section verifies one exact instance per boundary in which the layer’s obstruction fires while every coarser layer is silent. The two absorbed generalizations supply the connective theory: general information structures (Section) give Layers 2–5 their common semantics and their ordering — viability is monotone under refinement, certificates erode as structures refine, and incomparable structures can disagree in either direction; exit-time values (Section) give Layers 1 and 3 their common currency — the guaranteed survival time, whose level sets are the viability kernels and whose comparison against a deadline is the timing certificate.

The discipline of Abaee 2026c, Section 7, is kept: the architecture unifies *form*, not *cause*; each layer’s obstruction has a different epistemic status; and the strongest claims remain the companions’, each at the precision its development supports (Section).

2. The five layers

Fix the framework of Abaee 2026c: augmented states, information states, response correspondences, and the recursive viable selector $\Gamma_N(B)$. Each layer is a region of that framework: a family of correspondences, the certificate families that witness their emptiness, and the companions’ results that establish the witnesses.

Layer 1: dynamics. At a single branch the intersection is trivial, and the correspondence empties on dynamics alone: the finite-time exit certificate under the Isaacs-type drift condition, the comparison-function refinement, the explicit bound $t_{\text{exit}} \leq q(x_0)/\varepsilon \leq a/\varepsilon$ (Abaee 2026a). Verified base instance (Section , E5): every revealed branch of the hidden-regime system is indefinitely viable under full information — Layer 1 obstructs only when the drift condition genuinely holds, never merely from information loss.

Layer 2: belief action. The static intersection object $\Gamma^B(B) = \bigcap_{z \in B} \Gamma(z)$ at the admissible, safe, and tube-safe choices of response: the common-action certificate with Farkas and $m+1$ witnesses, the admissibility register, the minimal epistemic-emptiness construction, the tube obstruction and its ladder (Abaee 2026a). Verified boundary instance: the two-floor belief has both branches individually safe with an empty common safe set.

Layer 3: timing. The blind-window predecessor correspondence and the guaranteed survival time σ^* : the delayed-information bound, the load-bearing class declaration, and Abaee 2026b’s finite programs, whose infeasibility certifies failure of every measurable information-adapted controller with post-observation recourse priced in. Verified boundary instance: the timing cells $z_0 \geq 2$ of the audited grid are one-step-safe (Layer 2 silent) yet timing-certified.

Layer 4: certification. The label-selection correspondence $\Lambda_K(O^{-1}(y))$: no exact observation-only certifier when a fibre crosses the safe-set boundary; the certainly-safe set as the maximal sound relaxation (Abaee 2026a, 2026c). A verdict layer, not a policy-viability obstruction. Verified boundary instance: the two-floor index, a static model in which Layers 1–3 have no content and the certification obstruction fires.

Layer 5: policy class. The declared-class correspondence $\Gamma(B; \Pi', \cdot)$ for a restriction $\Pi' \subset \Pi$: the certainty-equivalence trap, which empties the kernel under an injective observation with Layers 1–4 silent (Abaee 2026a, Corollary 1 of 2026c). Verified boundary instance: the biased-index instance, where the corrected law is a witness for the unrestricted class and every certainty-equivalence law has drift at least 21/100 on the band.

3. Layer 2–5 on general information structures (D2)

The companions’ exact theorems assume exogenous information — history partitions of the hidden mode, independent of state, control, and disturbance. The selector framework needs less, and this section records how much less, exactly.

Definition 1 (information structure). An information structure on the augmented state space is a map $f : Z \rightarrow \mathcal{L}$ to a label space: f partitions Z into cells $f^{-1}(\ell)$, and f

is the observation the regulator reads when acting. f refines g if same f -label implies same g -label. Semantics: read-then-act — the regulator observes the current cell and selects one action per cell (states sharing a cell are indistinguishable and share a response; this is the common-action condition within cells); successors re-partition and the recursion continues.

Blind is the one-cell structure; full information is the singleton structure; the companions’ delayed model is blind up to a deadline and full after. Layer 2’s common-action certificate is the blind case; every certificate family of Sections 2–5 of Abaee 2026a is a statement about cells of some structure.

Proposition 1 (refinement monotonicity). Viability of a belief is monotone under refinement: if f refines g , then every f -viable belief is g -viable, and every g -certificate (emptiness witness) remains valid at f . Along any refinement chain the verdict changes at most from nonviable to viable; certificates erode precisely as the bridge’s identity $\Gamma_{\text{coarse}} \geq \Gamma_{\text{fine}}$ (Abaee 2026b) predicts on the viability side.

Proof sketch. A per-cell selection for the finer structure restricts to a per-cell selection for the coarser one by merging cells (the merged cell acts by one of the fine cells’ actions and is safe on the union because the coarse cell is partitioned by the fine one); conversely an emptiness witness for the coarser common-action problem applies verbatim within any fine cell it saturates. $\square$

Proposition 2 (the lattice matters). Refinement monotonicity does not extend off the refinement order: two incomparable structures can receive opposite verdicts on the same belief. Verified instance (Section , E1): on the two-patch audit system, the belief $\{(1, 2), (2, 1)\}$ is non-viable under the aggregate reading $y = z_1 + z_2$ and viable under the coordinate reading z_1 alone — the coordinate structure distinguishes what the aggregate merges, the aggregate saturates no coordinate cell, and neither refines the other. On the same system the refinement chain blind $<$ aggregate $<$ full is verdict-monotone with strict improvement exactly at the full refinement, and refinement strictly erodes the certificates: over all safe pairs, 28 beliefs are viable under full information against 26 under the aggregate reading (E2) — the erosion identity’s exact viability-side count.

Scope. Beyond exogenous structures — control-dependent observations, active sensing — the read-then-act semantics itself is no longer an exact representation of the policy class, and only sound statements survive: certificates for the exogenous relaxation remain valid upper envelopes of the true policy class, in the same soundness-only mode as the companion bridge’s oracle-recourse extension (Abaee 2026b) and the measurable-selection qualification of Abaee 2026c. This is the same delimitation the bridge states, kept verbatim at the architecture level.

4. Exit-time value functions (D3)

Layers 1 and 3 measure the same quantity from two sides.

Definition 2 (exit-time values). For an information state B , a declared class Π_B , and the constraint function q with $\mathcal{V} = \{q \geq 0\}$, the guaranteed survival time is

$$T^*(B) = \sup_{\pi \in \Pi_B} \inf_{z_0 \in B, d(\cdot)} \sup\{t : q(x(t)) \geq 0\},$$

with the full-information value at singletons written T_{full}^* and the blind-window value, for the delayed model, the companions' σ^* .

Proposition 3 (level sets are kernels). For every horizon T , $\{B : T^*(B) \geq T\} = \mathcal{W}_T$: the level sets of the exit-time value are exactly the finite-horizon viability kernels, and Layer 1's exit certificate is the statement $T^*(x_0) < \infty$ witnessed by a drift bound, while Layer 3's timing certificate is exactly $T^*(B_0) < T_{\text{obs}}$ — the value compared against the observation deadline. On the audited classes the identities hold cell by cell: on the hidden-regime grid, $\sigma^*(z_0) = z_0 - 1$ and $\{\sigma^* \geq T\} = \{z_0 \geq 1 + T\}$ for every audited deadline (48 cells); on the companion bridge's contraction instance, $\sigma^*(z_0) = \max\{k : z_0 \geq (\frac{10}{9})^k\}$, and the level sets coincide with the finite program's feasibility thresholds $\{z_0 \geq (\frac{10}{9})^K\}$ — the bridge's linear programs compute, exactly, the level sets of the exit-time value.

Proof sketch. $T^*(B) \geq T$ says some policy keeps every compatible trajectory in $\mathcal{V}$ through T , which is $\mathcal{W}_T$ membership by Theorem 1 of Abaee 2026c; the two families of computed identities are re-derivations in exact arithmetic (Section , E3–E4). The Hamilton–Jacobi–Isaacs reading of T^* — its viscosity characterization and comparison-function bounds — is available in the literature and is not redeveloped here; the programme's contribution is the level-set identification with the certificate layer. $\square$

Remark 1 (stochastic reading). Under genuinely stochastic kernels the exit-time value's expectation counterpart is the belief-state safety value of Abaee 2026d, and the deficit $1 - V_k$ is the stochastic analogue of $T_{\text{obs}} - T^*$: certificates read as deficit lower bounds there, as deadline comparisons here. The two readings coincide at the deterministic degeneration.

5. Layer separation

The architecture's content is at its boundaries; Table 1 lists the verified instance at each, each an existing companion instance re-classified by layer.

6. Verified record

`successor_five_layer_v1_verify.py` (standard library; exact rational and integer arithmetic; deterministic) carries seven check families, all passing: **E1** the information lattice on the two-patch system — blind, aggregate, z_1 -only, z_2 -only, and full verdicts (N, N, V, V, V) on both audited beliefs, the refinement chain verdict-monotone with strict improvement at full, and aggregate versus z_1 -only incomparable with opposite verdicts; **E2** erosion — 28 viable pairs under full against 26 under the aggregate reading; **E3** the hidden-regime level-set identity on all 48 cells; **E4** the contraction instance's exit-time value $\max\{k : z_0 \geq (\frac{10}{9})^k\}$ at exact rational powers and neighbors, with level sets equal to the bridge's feasibility thresholds; **E5** full-information branch

viability ($T^* = \infty$ on every revealed branch); **E6** the four boundary instances of Table 1. Two defects were found and fixed during development, both in the verification harness itself: an initial draft gave every information structure blind semantics (one common action per belief) instead of read-then-act per-cell semantics — the crash it caused is what surfaced the error — and a refinement predicate with an inverted implication; the manuscript's statements were unchanged, the harness was corrected before any record was archived.

7. What the architecture does not claim

The layers are a presentation discipline over the companions' results, not new mechanisms: every certificate cited here is proved in Abaee 2026a–d, and the new content of this paper is the two absorbed generalizations at their verified instances (Propositions 1 and 2, and the level-set identities of Proposition 3) together with the separation table. The layers do not rank importance: Layer 1 is not epistemic, Layer 4 is a certification limit rather than a policy-viability obstruction, and Layer 5 can operate under perfect observation. The strongest claim the architecture supports is the one Abaee 2026c already states, now with its two generalizations in place: observation-based viability requires a common recursively safe selector from information states to admissible responses; the certificates are checkable sufficient emptiness conditions, exact in finite-horizon finite models, sound local exclusion tests elsewhere, organized by the resource whose absence each detects.

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Declarations

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boundary	instance	coarser layers silent because	firing layer's certificate
L2 over L1	two-floor belief	each branch individually safe (drift nonnegative under its own action)	common safe set empty
L3 over L2	timing cells $z_0 \geq 2$, $T_{\text{obs}} > z_0 - 1$	one-step common action safe ($\sigma^* \geq 1$)	$\sigma^*(z_0) < T_{\text{obs}}$
L4 over L1–3	two-floor index (static)	no dynamics: L1–3 vacuous	fibre crosses the boundary: no exact certifier
L5 over L1–4	certainty-equivalence instance	injective observation (certifier exists); corrected law viable	every CE law: drift $\geq 21/100$, class correspondence empty

Table 1: One verified instance per layer boundary: the layer's obstruction fires while every coarser layer is silent (exact values in Section , E6). Together with the base layer these are the five layers of the architecture; no layer's instances are subsumed by a coarser one.